

Target-Data Access and Aggregation Effects in Cross-Subject EEG Evaluation

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Abstract—Cross-subject EEG decoders are compared across protocols that use the target subject differently, and the gaps between them are read as the value of that access—a reading that requires everything else to be matched. We divide each held-out subject’s trials once into an available half and a reserved test half, score every condition on the same reserved half, and cross two procedures, Euclidean Alignment and supervised training on the available half, holding the source partition, the per-epoch sample budget, the target sampling rate and the scored trials fixed and verified against the realized indices. On three motor-imagery datasets spanning 9 to 105 subjects, and under one stated training policy, supervision helps on all three and alignment on two, but their ordering reverses—supervision leads alignment by 7.9 pp on BCI IV 2a and trails it by 4.2 pp on PhysionetMI—while no interaction replicates. Fixing the batch share does not fix exposure—each calibration trial is replayed 1.3 to 16.6 times per epoch depending on how many exist—but capping the calibration set within a dataset moves the effect by only 2 of the 12 pp involved, so the reversal is not a budget artifact. A fixed-prediction control that only regroups identical predictions behaves oppositely: the between-subject spread exceeds what random groups of the same size give by 6.2, 2.4 and 1.8 times, and a conditionally binomial correction attributes 2%, 17% and 29% of the observed variance to finite trials. That changes a reading of these data—raw spreads rank PhysionetMI as the more heterogeneous cohort, corrected ones do not—so a between-subject standard deviation needs the trials it came from.

I. INTRODUCTION

Cross-subject EEG decoders are compared across papers whose evaluation protocols use the target subject differently. Some use none of it, some use its unlabelled signals to align feature distributions, some fine-tune on a labelled calibration block. The labels attached to these protocols—zero-calibration, unsupervised transfer, subject-adaptive—are coarse, and the accuracy differences between them are routinely read as the value of the extra access.

That reading requires the comparison to hold everything else constant, and published comparisons rarely can. Adding a calibration block lengthens an epoch, so the adapted model also gets more optimizer steps; the target’s share of a batch is whatever the cohort size implies; and if the block supplies the early-stopping signal, model selection has changed too. None of this is visible in a results table, and all of it moves with the effect being claimed. Our own earlier work tracked those

quantities rather than the procedure; Section II-C reports what replaying it shows.

This paper asks a narrower question that can be answered cleanly. **On a fixed target test set, what does each way of using the target subject’s available trials buy, once the source partition, the per-epoch sample budget, the early-stopping rule and the target sampling rate are held constant?** Each held-out subject’s trials are divided once into an available half and a reserved test half, and every condition is scored on the same reserved half. The available half is used in one of four ways: not at all, by Euclidean Alignment, by supervised training, or by both. This crosses two *procedures*, not two independent kinds of access—supervised training reads the available trials’ signals as well as their labels—so the grid separates what each procedure adds on top of the other, not the value of signals against labels.

Related work. Partitioning and reporting choices are known to move BCI comparisons [1], [2], and MOABB standardizes the interface but not the accounting [3]: two pipelines can share a loader and a split while giving the target subject different influence over training, normalization and checkpoint choice. Euclidean Alignment reads the evaluated subject’s trials but none of their labels and is routinely called zero-calibration [4], while calibration-block fine-tuning consumes labels outright; compact convolutional references [5], [6] carry both. Leakage drives irreproducible findings [7] and small-sample neuroimaging yields error bars wider than reported [8], but that work is diagnostic, whereas choosing between protocols needs what a procedure is worth once everything else is matched.

A second question concerns how the result is summarized. A cross-subject paper reports a spread as well as a mean, and the spread is read as between-subject heterogeneity. That holds only if the aggregation unit is the subject. We test it with a control that retrains nothing: the same per-trial predictions are aggregated by true subject and by random groups of the same sizes, so any difference is attributable to the grouping alone.

Contributions. (i) A controlled 2×2 of alignment against target supervision on three motor-imagery datasets spanning 9 to 105 subjects, with the source partition, the epoch and the scored trials pinned and checked against the realized indices, and with the exposure quantities that *cannot* simultaneously

be pinned reported instead. (ii) A fixed-prediction regrouping benchmark, and a matching variance estimate, for how much of a reported between-subject spread the aggregation unit accounts for. (iii) An audit of what an uncontrolled version of the same comparison reports instead. Code, logs and per-trial predictions: <https://github.com/yo666666yo/target-data-access-eeg>. The split, the resampling and alignment itself are standard; what is new is the set of controls under which the comparison is made, and the account of which cannot hold at once.

II. CONDITIONS AND CONTROLS

A. Splitting the target subject

We evaluate leave-one-subject-out on three motor-imagery datasets [3], [9]: the official BCI Competition IV 2a release, read from the competition archives (9 subjects, 22 channels, four classes, band-pass 8–32 Hz over the four seconds after the cue, EOG dropped by the official channel contract, sessions T and E concatenated in that order); BNCI2014-002 (14 subjects, 15 channels, right hand versus feet); and PhysionetMI (105 usable subjects, 64 channels, four classes after dropping *rest*; subjects 88, 89, 92 and 100 excluded for non-standard sampling). The latter two are loaded through MOABB’s MotorImagery paradigm at its default 8–32 Hz, each over its own declared trial interval. The decoder is our own reimplementation of EEGNet-8,2 [5], without max-norm constraints and not validated against a reference implementation, trained with Adam (10^{-3} , weight decay 10^{-4}), batch 64, ≤ 50 epochs, step decay ($\times 0.5$ every 15) with patience 10 and class-balanced loss, features standardized by one scaler fitted on source training trials.

Each held-out subject’s trials are divided once, stratified by class, into an *available* half C_s and a *reserved test* half T_s . That 50/50 division is the declared target-data budget: 288, 80 and 45 trials per half. It is seeded per subject, so no condition can change which trials are reserved. T_s is the only thing any condition is scored on, and no condition touches it.

B. Two procedures, four conditions

Figure 1 summarizes the design. The two procedures are not independent kinds of access: supervised training on C_s reads those trials’ signals as well as their labels. The grid crosses procedures, not permissions.

Alignment uses signals and no labels. Euclidean Alignment [4] whitens a subject’s trials by $\bar{R}^{-1/2}$, where $\bar{R}$ is the mean over that subject’s trials of the sample spatial covariance XX^T/T , computed on the band-passed epochs with no shrinkage and inverted through an eigendecomposition whose eigenvalues are floored at 10^{-10} to keep a rank-deficient epoch from blowing up. Source subjects are whitened by a reference estimated from all of their trials, which is legitimate because all of them are available. The held-out subject’s reference is estimated from C_s alone and then frozen; T_s is transformed by it but never contributes to it. The switch also changes how every source subject is preprocessed, so the measured effect is that of the whole alignment pipeline. *Target supervision* adds C_s to the training set with its labels attached.

The four conditions are neither procedure (SRC), alignment only (EA), supervision only (SUP), and both (EA+SUP). Two further arms sit off the grid, each changing a third thing: *SEL* picks the checkpoint on C_s rather than the source validation set, leaving training set, initialization and trajectory unchanged but swapping a selection set of very different size; *POOL* adds C_s to the same scaler’s fit set. T_s enters no scaler.

C. What is held fixed

Three quantities are pinned exactly. **The source partition:** the stratified 80/20 split supplying training and selection data is derived from the outer pool alone and is the same index set in all four conditions; validation is source-only throughout, so target trials never reach the selection set. **The epoch:** every condition draws, with replacement, as many samples per epoch as the source training set holds (3686, 1664 and 7486). **The scored object:** T_s is identical across conditions, trial for trial. A fourth, **the target sampling rate**, is fixed in expectation: a plain shuffled loader over the same pool would give the target 7.2%, 4.6% and 0.6% of a batch—a 12-fold range set by cohort size—so the supervised arms use a weighted sampler with a declared expected share of 0.10, realized at 0.096–0.109.

Two quantities this does *not* equalize, which we report rather than claim. Because the epoch is a fixed total, a supervised arm draws 10% fewer source samples per epoch than SRC; and because each arm early-stops on its own curve, realized updates differ—taking fold-averaged totals per arm and dividing the spread by the smallest, by 10% on 2a, 3% on PhysionetMI and 23% on BNCI2014-002. Exposure is the sharper problem: a declared share w implies $wS/|C_s|$ expected draws of each target trial per epoch—1.3, 2.1 and 16.6 here, since $|C_s|$ differs 6.4-fold. Matching repetition instead would need shares of 0.156, 0.096 and 0.012; matching both would need source exposure to vary 13-fold. Any such comparison fixes two of these three and lets the third move, which is why we vary $|C_s|$ directly in Section III.

We state these because an earlier version of this study enforced none of them. Replaying its fold construction shows a supervised arm on a re-drawn source partition, with target trials in its own validation set and batch shares of 6.1, 3.8 and 0.5%—a gradient its cross-dataset ordering followed—a selection arm trained on 25% more source trials than its reference, and an aligned arm whose whitener was fitted on $C_s \cup T_s$. None is visible in a results table, so the controls are checked against the realized indices by a script run before the experiments.

D. Aggregation unit and statistics

A cross-subject spread is read as between-subject heterogeneity, which requires the aggregation unit to be the subject. We test that with a control that retrains nothing: the per-trial predictions on T_s are held fixed and only the grouping changes, first by true subject, then into random groups of the same sizes, 2000 times, on SRC and SUP. Trial-weighted accuracy is identical either way, and subjects contribute equal trial counts on 2a and BNCI2014-002 and near-equal ones on PhysionetMI (45 for 104 subjects, 44 for one), so the subject macro-average

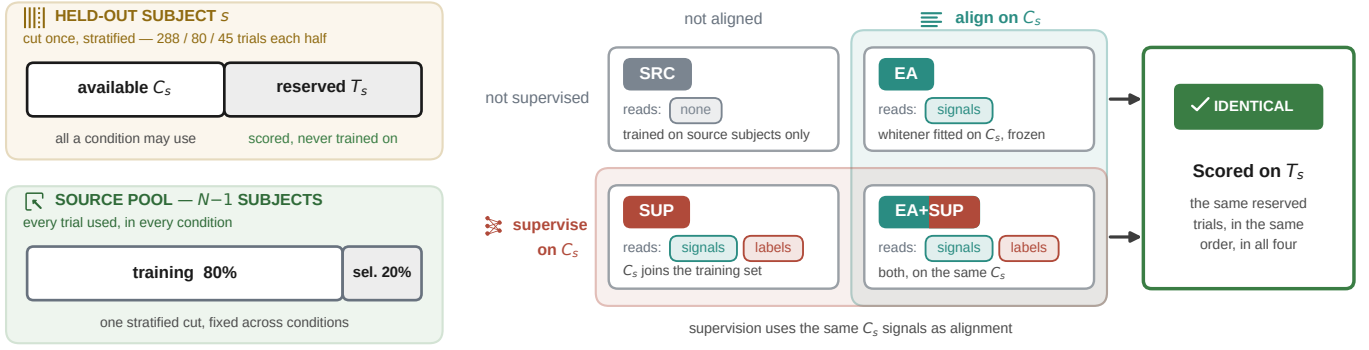


Fig. 1. Left: the two divisions of the data, each made once and identical in every condition. Right: the four conditions as a 2×2 over the two procedures, with what each reads from the held-out subject’s available half C_s . Supervised training necessarily reads those trials’ signals as well as their labels, so the grid crosses two *procedures*, not two independent kinds of access. Everything else is held fixed: the source training set, the selection set, the samples drawn per epoch and the expected target share of a batch, and all four are scored on the same reserved trials in the same order.

and the trial-weighted average coincide up to that single trial and any difference in spread belongs to the grouping alone. The null is not a new phenomenon—drawing m of M correctness records without replacement gives group means of variance $(1 - m/M)S^2/m$, which our resampled null reproduces—but it says how much of a *reported* spread that accounts for, where m ranges from 45 to 288.

Each subject contributes one number per condition, averaged over three runs. Those runs vary the target split and the initialization together, so they estimate neither alone; within a run all conditions share both, which is what the pairing requires. Differences are paired within subject and resampled over subjects, with one set of bootstrap indices drawn per dataset and reused for every condition (10^4 replicates, percentile intervals). p -values come from a two-sided sign-flip permutation test over the five conditions of a dataset, Holm-corrected within that family, with a paired t -test as a sensitivity check. That test saturates on a small cohort—on nine subjects it cannot return less than $2/2^9 = 0.0039$ —so we report the number of subjects helped beside it. Spread comparisons are ratios of standard deviations, not variances.

Because leave-one-subject-out folds share most of their training pool, the subjects are not independent replicates, and with nine of them on 2a the reference distribution is coarse. We therefore read a Holm-adjusted p below 0.05 as a difference resolved by this repetition structure—these subjects, these runs, this training policy—and not as inference about a subject population, and we use that as the single rule for which differences we state. Intervals give magnitude only: they carry no multiplicity adjustment, and in a Gaussian simulation at $n = 9$ our percentile intervals covered 89% rather than 95%, so an interval excluding zero beside a non-significant adjusted p is not evidence of an effect.

III. RESULTS

A. What each procedure buys

Table I gives the source-only references and the effects. Chance differs across the three datasets, so the reference accuracies are not comparable to one another; every claim below

is a within-subject difference against an arm’s own reference. Supervision rejects on all three datasets and alignment on two of three, by amounts that do not carry across them, and the ordering between the two reverses.

Alignment is worth +1.58 pp on 2a, where it does not survive correction (7 of 9 subjects, Holm $p = 0.18$), against +8.07 and +9.12 pp on BNCI2014-002 and PhysionetMI, where it does (12/14 and 93/105, $p \leq 0.005$). Target supervision runs the other way: +9.45 pp on 2a, helping every subject, falling to +6.52 and +4.89 pp on the larger cohorts. Comparing the two directly—exploratory, prompted by this ordering rather than specified in advance—supervision leads alignment by +7.87 pp [+4.95, +11.03] on 2a, is indistinguishable from it on BNCI2014-002 (−1.55 [−4.46, +0.89]), and trails it by −4.23 pp [−5.84, −2.64] on PhysionetMI. Both endpoints reject under the decision rule—Holm-adjusted within these three tests, $p = 0.008$ and $p < 0.001$ —so the two datasets are not scattered around a common value. Being exploratory, this is an observed reversal in these runs, not a confirmed ordering.

The calibration budget is confounded with the dataset, $|C_s|$ differing 6.4-fold, so we re-ran the arms that consume it with $|C_s|$ capped by a nested stratified subset and T_s untouched. Shrinking it moves the effect monotonically but only a little: at a common budget of 45 trials the three datasets read +5.88, −2.62 and −4.23 pp, so the ordering holds and the budget accounts for about 2 of the 12 pp separating 2a from PhysionetMI. The full curve is in the supplement.

B. Does it survive a different decoder, or a chronological split?

Two things could make these effects artefacts of our setup rather than of the protocol. The random split interleaves the halves in time, so they share drift and attention state; BCI IV 2a’s two sessions let the same 288/288 budget be partitioned chronologically, and over five runs the effects barely move (SUP − EA +7.75 pp [+5.54, +10.29] against +7.87), making the contrast cross-session there. And the decoder is our own reimplementation, so we repeated the grid with braindecode’s EEGNetv4 (0.8.1, matched hyper-parameters) swapped in: source-only accuracy moves −0.81, +0.27 and −0.55 pp, so

TABLE I

EFFECT OF EACH PROCEDURE AGAINST THE SOURCE-ONLY REFERENCE UNDER THE CONTROLS OF SECTION II-C. DIFFERENCES ARE PAIRED WITHIN SUBJECT AND AVERAGED OVER THREE RUNS; THE PER-RUN COLUMN GIVES THE SAME DIFFERENCE WITHIN EACH RUN. INTERVALS ARE DESCRIPTIVE PERCENTILE BOOTSTRAP OVER SUBJECTS, NOT MULTIPLICITY-ADJUSTED; p IS HOLM-CORRECTED WITHIN EACH DATASET AND IS FOLLOWED BY THE NUMBER OF SUBJECTS HELPED, WHICH DISTINGUISHES ARMS AT THE PERMUTATION FLOOR. SD IS THE BETWEEN-SUBJECT STANDARD DEVIATION OF RUN-AVERAGED PER-SUBJECT ACCURACY—A DIFFERENT POOLING ORDER FROM SECTION III, WHICH POOLS VARIANCES WITHIN RUNS. SEL AND POOL SIT OFF THE GRID AND ARE QUOTED IN THE TEXT.

Condition	Uses C_s to	Δ accuracy (pp) [95% CI]	per run (pp)	subject SD	SD ratio [95% CI]	p (improved)
<i>BCI IV 2a</i> ($n = 9$ subjects; source-only accuracy 0.450, between-subject SD 0.175)						
EA	align	+1.58 [+0.17, +2.98]	+0.7 +2.1 +1.9	0.180	1.03 [0.94, 1.13]	0.176 (7/9)
SUP	supervise	+9.45 [+5.88, +12.81]	+9.0 +9.1 +10.3	0.196	1.12 [0.88, 1.36]	0.019 (9/9)
EA+SUP	align + supervise	+13.66 [+9.53, +17.44]	+12.2 +13.5 +15.3	0.199	1.14 [0.84, 1.36]	0.019 (9/9)
<i>BNCI2014-002</i> ($n = 14$ subjects; source-only accuracy 0.687, between-subject SD 0.115)						
EA	align	+8.07 [+4.37, +11.96]	+6.3 +7.5 +10.4	0.120	1.04 [0.78, 1.49]	0.005 (12/14)
SUP	supervise	+6.52 [+3.18, +10.27]	+5.9 +3.8 +9.8	0.136	1.18 [0.95, 1.64]	0.005 (12/14)
EA+SUP	align + supervise	+11.13 [+6.34, +16.16]	+10.8 +10.4 +12.2	0.118	1.02 [0.68, 1.59]	0.002 (13/14)
<i>PhysionetMI</i> ($n = 105$ subjects; source-only accuracy 0.444, between-subject SD 0.124)						
EA	align	+9.12 [+7.40, +10.81]	+10.9 +7.0 +9.5	0.151	1.22 [1.09, 1.37]	<0.001 (93/105)
SUP	supervise	+4.89 [+3.41, +6.41]	+5.5 +3.7 +5.5	0.159	1.28 [1.16, 1.42]	<0.001 (72/105)
EA+SUP	align + supervise	+12.68 [+11.04, +14.35]	+13.9 +10.5 +13.7	0.159	1.28 [1.16, 1.43]	<0.001 (99/105)

the low reference belongs to zero-calibration evaluation rather than our code, and the reversal reproduces at +8.27 pp on 2a and −4.65 on PhysionetMI. EA and SUP reproduce wherever they reject; SEL on BNCI2014-002 does not. These runs and the capped-budget ones carry their own source snapshots.

Whether the two compose is undetermined: the interaction $(EA+SUP - EA) - (SUP - SRC)$ is +2.62, −3.45 and −1.33 pp, none rejects, and per run its sign is consistent within each dataset but not between them. Its sign may be dataset-dependent, and none of this is evidence of additivity.

C. The aggregation unit

Holding the predictions fixed and changing only the grouping, the resampled null matches the finite-population prediction to within 0.0013 and tracks the trials per group, though it also depends on the pooled accuracy and the group count. With 288 test trials each on 2a it is 0.029 against 0.177 by true subject, a factor of 6.2, falling to 2.4 with 80 trials and 1.8 with 45; the supervised arm behaves likewise (6.99, 2.96, 2.23), though not at the same size, so the gap is not independent of training.

That null is a benchmark, not a decomposition: shuffling removes the subject structure along with the trial noise. Estimating the two components separately instead, under a conditionally binomial model, puts finite trials at 2%, 17% and 29% of the observed between-subject variance and leaves standard deviations of 0.174, 0.111 and 0.113 against raw values of 0.177, 0.122 and 0.134; the estimator, its assumptions and the per-run terms are in the supplement. Raw figures rank PhysionetMI as the more heterogeneous cohort and corrected ones do not, though that gap is unresolved (+0.002 [−0.031, +0.024]). The control itself establishes only that spread exceeds the benchmark everywhere ($p < 0.001$).

D. The two arms off the grid

Choosing the checkpoint on C_s can cost accuracy: SEL is −1.69 pp [−2.80, −0.55] on PhysionetMI, helping 37 of 105

subjects, surviving correction ($p = 0.009$) and reproducing under the reference decoder (−1.81)—and that is where the selection set is smallest, 45 trials against the 1872 it replaces. Its positive values elsewhere do not reproduce, so we claim only this direction. No POOL arm rejects.

IV. DISCUSSION

With the source partition, the epoch and the scored trials fixed, what target data buys is not portable: the ordering of the two procedures reverses across datasets and target-subject selection changes sign. The aggregation unit behaves oppositely: its benchmark is orderly and, on these datasets, tracks the trials per group.

Some readings these numbers do not support. The reference accuracies sit below those usually quoted, but standardization, budget, schedule, alignment and a reference decoder each failed to account for the gap, so it is not our implementation. A large alignment gain is likewise no evidence of a violation: admissibility turns on whether the claimed deployment permits reading the target’s unlabelled signals. Nor is the supervised effect the value of labels. A fixed batch share replays each PhysionetMI target trial 16.6 times per epoch against 1.3 on 2a, which made overfit to a small, repeatedly drawn C_s a candidate explanation for the reversal; capping $|C_s|$ rules that out, moving the effect by about 2 of the 12 pp involved. What remains is a property of the datasets, which we do not resolve further. SEL’s harm is a separate route: it never trains on C_s , only picks a checkpoint there, so what is at stake is selection noise on 45 validation trials rather than training overfit—a distinction these runs do not resolve.

Limitations. Two decoders, one alignment method, one sampling rate, one 50/50 budget; the chronological split was available only on 2a. LOSO folds share training pools, so intervals describe these runs, not a population. Runs vary split and initialization together.

COMPLIANCE WITH ETHICAL STANDARDS

This work is a secondary analysis of three public motor-imagery EEG corpora—BCI Competition IV dataset 2a, BNCI2014-002 and the PhysioNet EEG Motor Movement/Imagery database—each released by its originators for open research use with participant consent and ethical approval obtained by the original investigators. No new data were recorded and no human participants were involved in this study, so no additional ethical approval was required. The authors declare no conflict of interest.

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